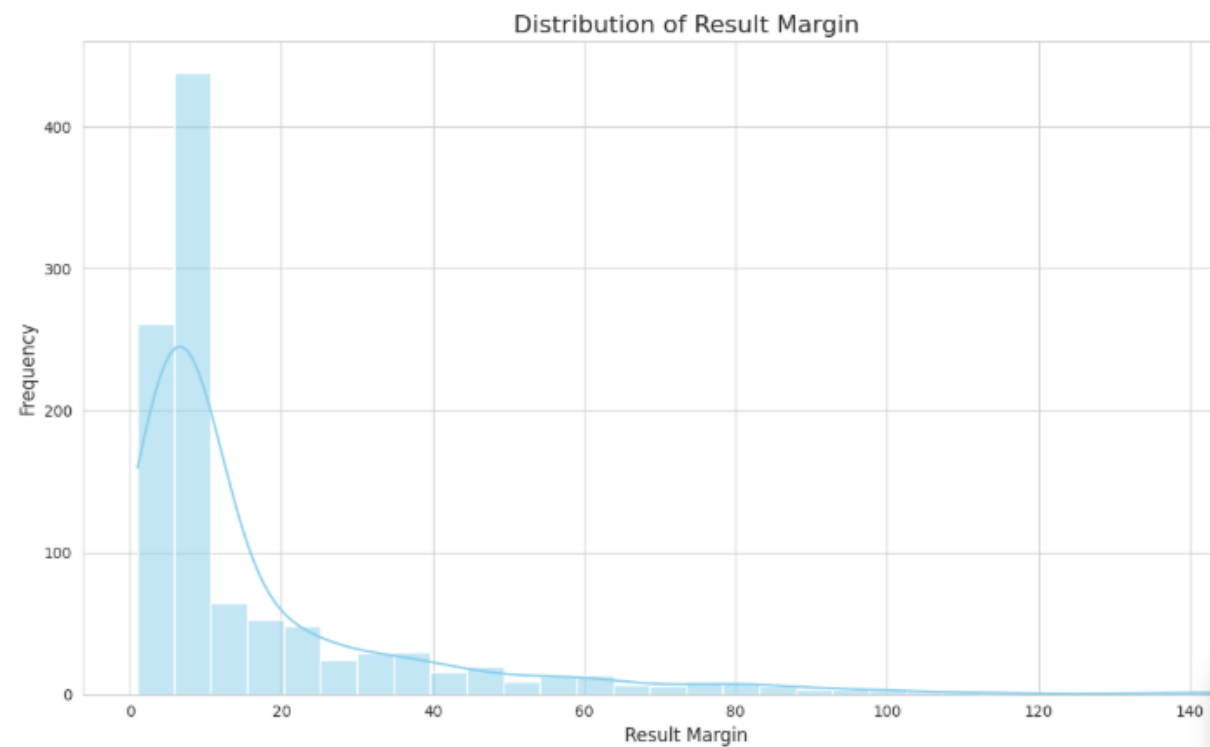
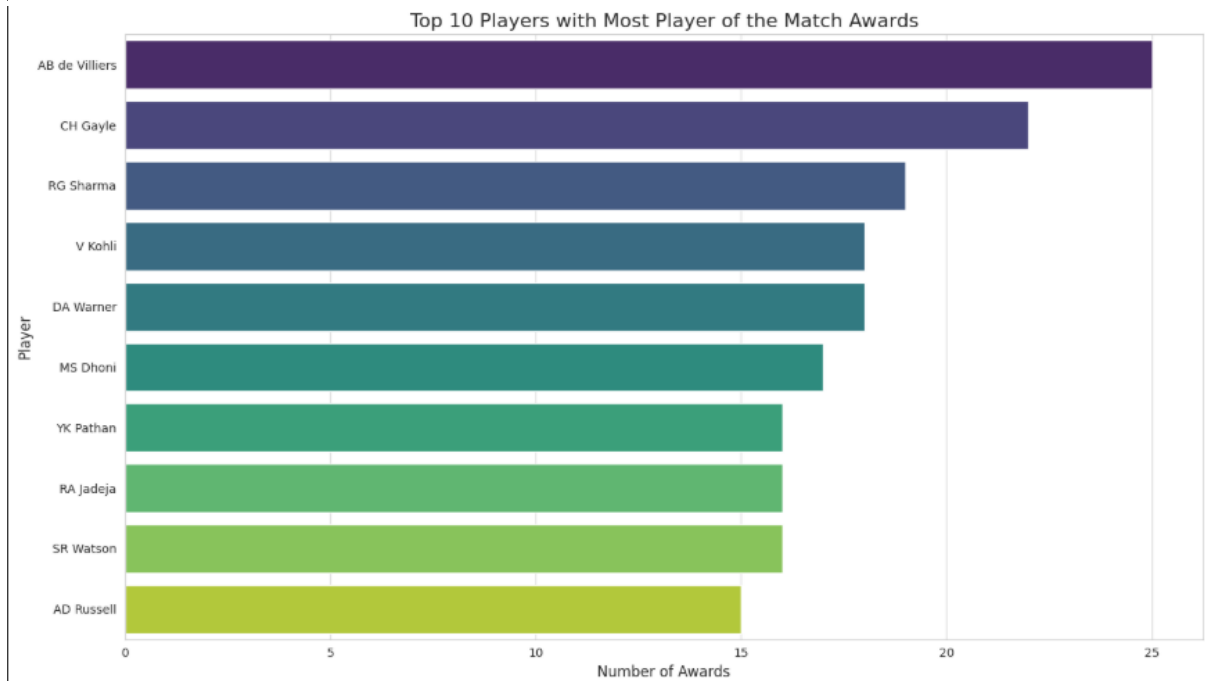
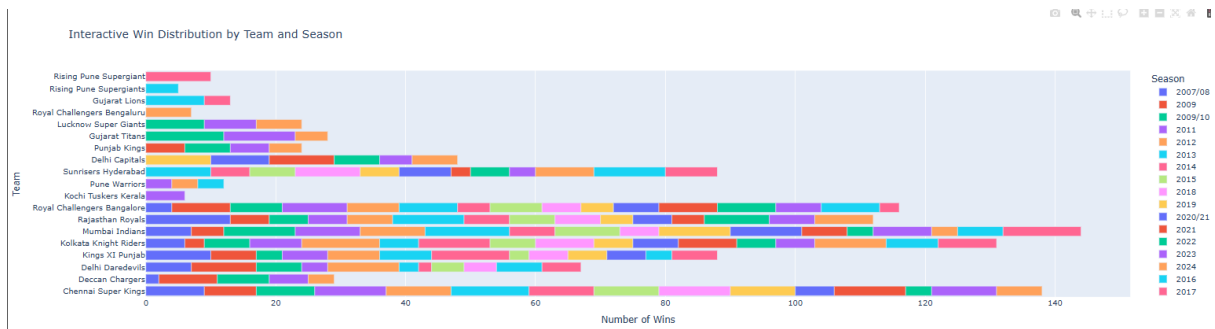
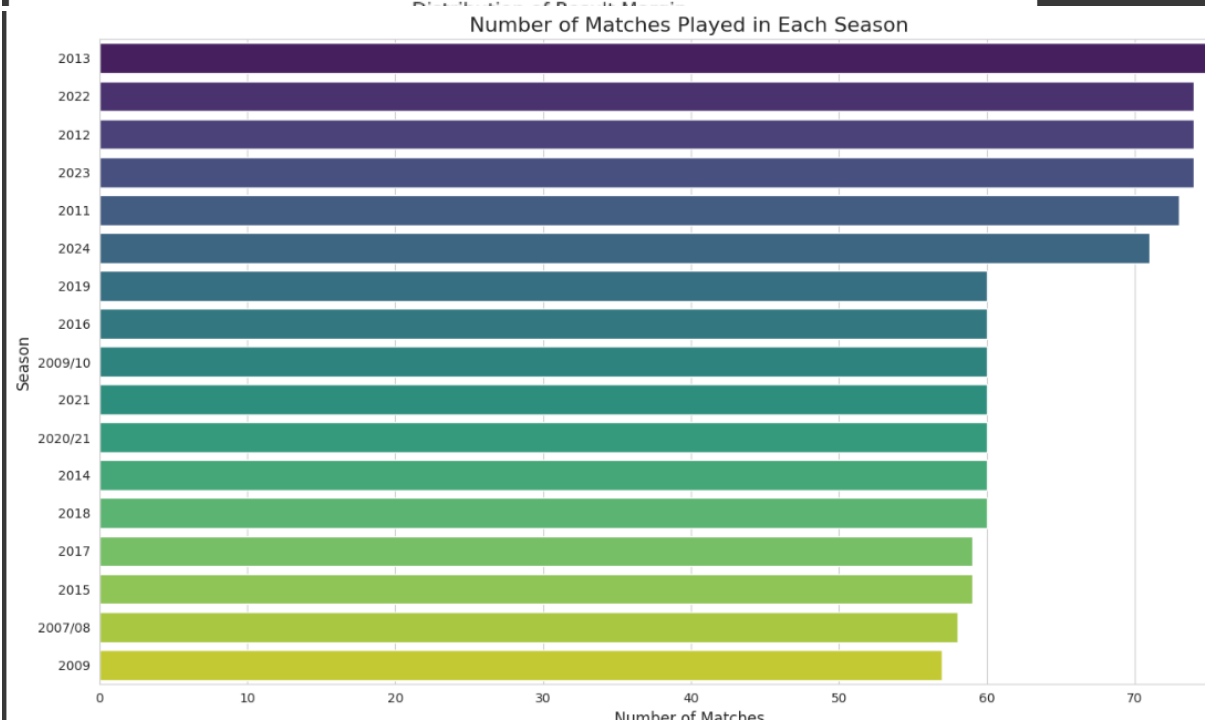
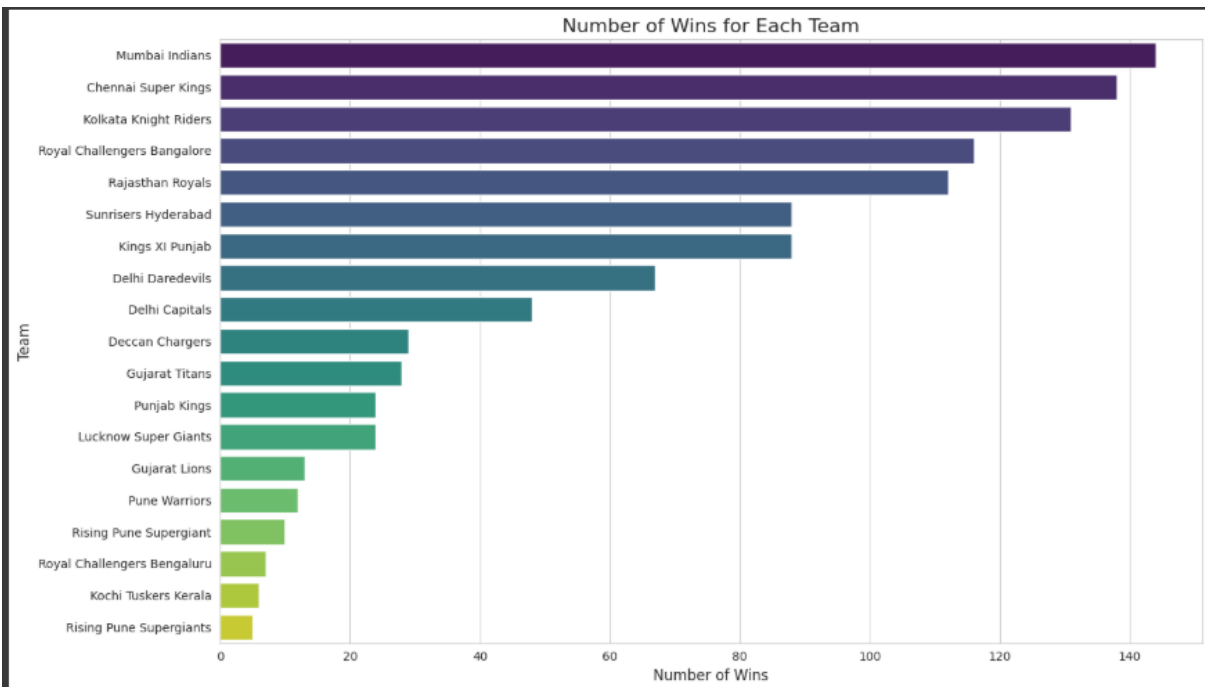
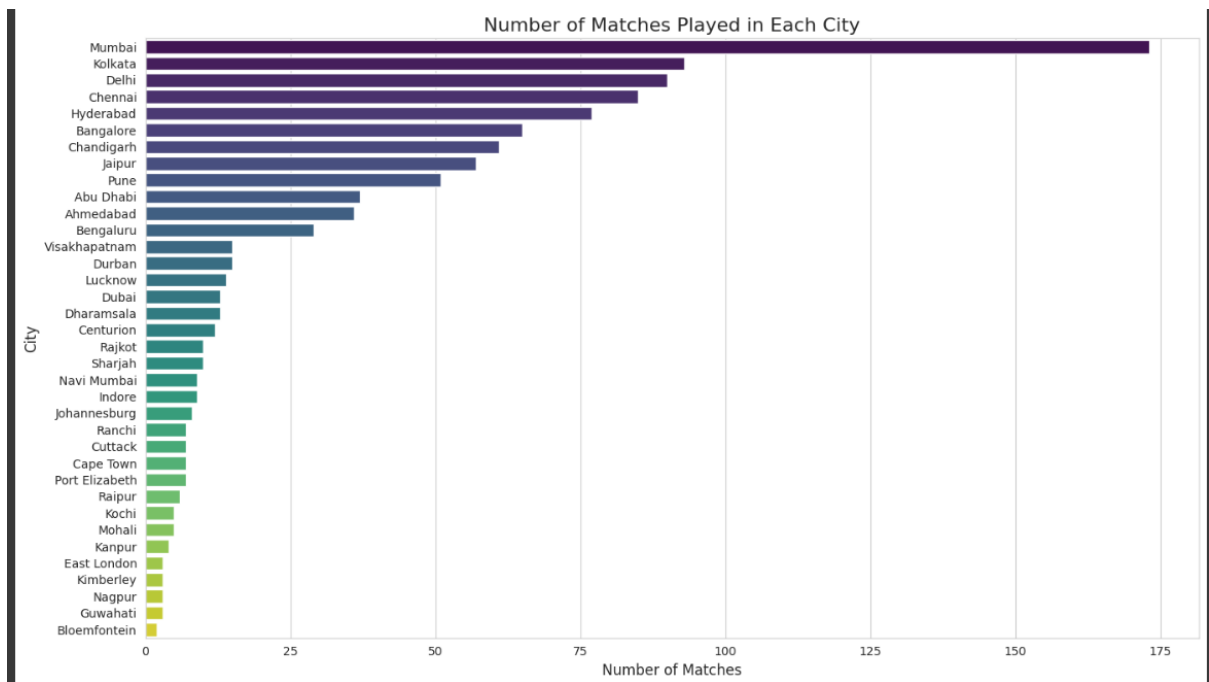








```
import zipfile

zip_file_path = "/content/archive (2).zip"
with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
    zip_ref.extractall()
```

```
df = pd.read_csv('matches.csv')
display(df.head())
```

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Set a visually appealing style for the plots
sns.set_style("whitegrid")

# 1. Bar plot showing the number of matches played in each city.
plt.figure(figsize=(14, 8)) # Increased figure size
sns.countplot(y='city', data=df, order=df['city'].value_counts().index, palette='viridis') # Changed color palette
plt.title('Number of Matches Played in Each City', fontsize=16) # Increased title font size
plt.xlabel('Number of Matches', fontsize=12) # Increased label font size
plt.ylabel('City', fontsize=12) # Increased label font size
plt.tight_layout()
plt.show()

# 2. Bar plot showing the number of matches played in each season.
plt.figure(figsize=(14, 8)) # Increased figure size
sns.countplot(y='season', data=df, order=df['season'].value_counts().index, palette='viridis') # Changed color palette
plt.title('Number of Matches Played in Each Season', fontsize=16) # Increased title font size
plt.xlabel('Number of Matches', fontsize=12) # Increased label font size
plt.ylabel('Season', fontsize=12) # Increased label font size
plt.tight_layout()
plt.show()

# 3. Bar plot showing the count of wins for each team.
plt.figure(figsize=(14, 8)) # Increased figure size
sns.countplot(y='winner', data=df, order=df['winner'].value_counts().index, palette='viridis') # Changed color palette
plt.title('Number of Wins for Each Team', fontsize=16) # Increased title font size
plt.xlabel('Number of Wins', fontsize=12) # Increased label font size
plt.ylabel('Team', fontsize=12) # Increased label font size
plt.tight_layout()
plt.show()

# 4. Histogram of the result_margin column
plt.figure(figsize=(12, 7)) # Increased figure size
sns.histplot(df['result_margin'].dropna(), bins=30, kde=True, color='skyblue') # Changed color
plt.title('Distribution of Result Margin', fontsize=16) # Increased title font size
plt.xlabel('Result Margin', fontsize=12) # Increased label font size
plt.ylabel('Frequency', fontsize=12) # Increased label font size
plt.tight_layout()
plt.show()

plt.figure(figsize=(14, 8)) # Increased figure size
top_players = df['player_of_match'].value_counts().nlargest(10)
sns.barplot(x=top_players.values, y=top_players.index, palette='viridis') # Changed color palette
plt.title('Top 10 Players with Most Player of the Match Awards', fontsize=16) # Increased title font size
plt.xlabel('Number of Awards', fontsize=12) # Increased label font size
plt.ylabel('Player', fontsize=12) # Increased label font size
plt.tight_layout()
plt.show()
```

```
import plotly.express as px

# Prepare data for Plotly
win_counts = df.groupby(['winner', 'season']).size().reset_index(name='wins')

# Create interactive bar plot using Plotly
fig = px.bar(win_counts,
             x='wins',
             y='winner',
             color='season',
             title='Interactive Win Distribution by Team and Season',
             labels={'wins': 'Number of Wins', 'winner': 'Team', 'season': 'Season'})

fig.show()
```

```
# Set a visually appealing style for the plots
sns.set_style("whitegrid")

# Analyze the impact of toss decision on win outcome
plt.figure(figsize=(12, 7)) # Increased figure size
sns.countplot(x='toss_decision', hue='winner', data=df, palette='viridis') # Changed color palette
plt.title('Impact of Toss Decision on Match Outcome', fontsize=16) # Increased title font size
plt.xlabel('Toss Decision', fontsize=12) # Increased label font size
plt.ylabel('Number of Wins', fontsize=12) # Increased label font size
plt.tight_layout()
plt.show()

# Analyze the impact of toss decision on win outcome - Filtered for 2024 season and specified colors
```

